



GUIDELINES FOR TRENCHING ACTIVITY IN PUNE CITY



**WORK IN
PROGRESS
INCONVENIENCE
REGRETTE**



GUIDELINES FOR TRENCHING ACTIVITY IN PUNE CITY

Booklet with the provisions in section 240 and 240 (2) of the
Maharashtra Mahanagarpalika act 1949 regarding road
trenching in Pune city



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Horizontal Directional Drilling Machine



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FOREWORD

Pune Municipal Corporation is committed to providing its citizens with the highest level of infrastructure facilities. The Road network forms a major part of the basic infrastructure constructed and maintained by the Municipal Corporation. The City of Pune has witnessed a burgeoning growth in the last decade, mainly due to the development of the IT sector in Pune. The residential and commercial markets of the city have also grown at an exponential rate and this has contributed to a steep rise in the vehicular population as compared to other cities in the country. The Municipal Corporation is also burdened with the responsibility of hosting a network of numerous utilities under the roads of the City, which are in a process of constant up-gradations. The upkeep of the road infrastructure within the available budget is a challenge before the Municipal machinery.

In Pune City, pipelines/cables of almost all utilities are laid underground, generally below the road surface which may include the carriage way or the footpaths. By virtue of increase in population and living standards, the demand for services, has not only just increased but the citizens, also expect better quality and prompt services from the Municipal Corporation and all other service providers. The services with pipelines or cables laid underground are basically Water supply, Drainage, Electric supply, Telecommunications, Gas line, CCTV etc. These services have caused great congestion below the road surface due to which laying of new pipeline or cables has become difficult and the occurrence of faults, requiring excavation for carrying out repairs, has increased. For laying of new pipelines/cables as well as for repairing faults, the concerned utility agency repeatedly seeks emergency permission from Road department. However, on completion of laying pipelines/cables or after the fault is repaired, it becomes the responsibility of the Corporation to bring the excavated surface back to its original status. To do this, the Corporation recovers the costs from the concerned utility, but it is observed that repairing does not bring the surface of the road or the footpath to its original status or in tune with the surroundings. Inconvenience is caused to the public, as the reinstatement and restoration cannot be done immediately. This is not only because of the technical deficiencies while carrying out the work, but also due to a number of other limitations like reinstatement of only excavated area, delay in reinstatement, communication gap between Utility agency and the Corporation or between various Departments of the Corporation. In case of new connections, fault repairs, checking of pressure by Water Works, and Sewerage Dept., as well as for cable fault and laying of cable for street light, etc. reinstatement is carried out departmentally, with the available material resources and at times it is not up-to the required standards.

Even though the necessary reinstatement charges are recovered from the concerned parties, on many occasions, due to the prevailing circumstances and practice, it becomes extremely difficult to maintain quality work, due to the physical constraints. To overcome this difficulty, the General body resolution No. 50dt.20/05/2014 has been issued. Subsequently, certain modification/ circulars are issued vide circular No. Road/4740 dated: 11/11/2014 and Road / 7006 dated 20/01/2014. Arising out of the recommendations of STAC (Roads) & to implement the General Body resolution effectively, the guidelines for carrying out trenching activity have become necessary. A Steering Committee has formed for co-ordination & communication between various departments of Pune Municipal Corporation and service providers /utility agencies. From the financial year 2016-17 a new trenching free technology will be made compulsory, barring exceptional cases, while granting road digging permissions.

As far as possible, all aforesaid modifications & circulars are now updated in this guidelines and the same are to be followed hence forth by all the concerned stakeholders.



Kunal Kumar
Municipal Commissioner
Pune Municipal Corporation



ADDRESS OF CHIEF ENGINEER FOR GUIDELINES OF TRENCHING ACTIVITY

It is observed that settlement or sinking of reinstated trenches for underground utilities along and across the roads in Pune City in monsoon is one of the major causes leading not only to poor riding quality but also consequent damage to surface and subsurface section of the roads. Settlement and water stagnation result into potholes, accidents and discomfort to the citizens and immense criticism to the Municipal Corporation.

During the Financial Year (2014-15) the permissions for excavation of trench for utilities, given by Pune Municipal Corporation ad-measured 329 km, out of total length of Pune Roads, which stands at 2200 km. Internal departments of Pune Municipal Corporation, have excavated trenches for their services which further add to the length of trenching in the city.

There are number of utility agencies, external as well as internal who have their services laid below Pune Roads and areas listed below:

No.	External agencies	Sr.	Water Supply
1.	MSEDCL	1.	Sewerage
2.	BSNL	2.	Storm Water Lines
3.	MNGL	3.	Electrical Works (street light & feeder pillar line for PMC services)
4.	MSEDCL Infra	4.	
5.	Tata Teleservices	5.	
6.	Tata Telecommunications	6.	
7.	Reliance JEO Infocom	7.	
8.	Reliance Infra	8.	
9.	Vodafone Ltd	9.	
10.	Idea Cellular	10.	
11.	Bharati Airtel Cos	11.	
12.	Other Private entities	12.	
13.	CCTV Network	13.	
14.	MSETCL	14.	

These agencies carry out digging/excavation of the road for repairs maintenance and augmentation of services. Earlier, there was no Apex Committee of Pune Municipal Corporation for monitoring of the trenching activities and hence there was no systematic planning or annual program for laying/ repair of various utilities etc. within the Pune Municipal Corporation departments or external utility service providers.

Hon. Municipal Commissioner has formed The Steering Committee for co-ordination & communication between various departments of PMC and service providers /utility agencies. Add. Municipal Commissioner (S), City Engineer of Pune Municipal Corporation, Chief Engineer (Road) , DMC (Zone no 1 to 4), Superintendent Engineer of Roads, Drainage, Water supply, Deputy Police Commissioner of Traffic, Pune City and the representatives of all external utility agencies /service providers are the members of the Steering Committee.

Trenches excavated by various departments of Pune Municipal Corporation for laying down internal utilities, are reinstated by the Road department or concerned Ward Offices. The external utility service provider deposit the amount for reinstatement work with Pune Municipal Corporation, based on a standard rate as decided upon by the General Body. The Road department of Pune Municipal Corporation, then, invites tenders for reinstatement of the utility crosscuts for such works. There is a certain provision in the budget for the reinstatement works. However, this amount is not commensurate with the total expenditure required during a particular year and hence there is a shortfall in the budget for reinstatement work of Pune Municipal Corporation. (Additionally, the cost of reinstatement for the works carried out by internal department of PMC is also not appropriated back to the Road department by that concerned department.)

The contractor, thus appointed by Road department executes the works under the direct supervision of the Junior Engineer of the Road department. The works are covered under defect liability period of one year from the date of completion of top asphalt layer.

SHORTFALLS /PROBLEMS AND SOLUTIONS

The main problem observed during such works is enforcing the different conditions laid down at various stages for trenching and the reinstatement. There are specific tender conditions for protection of trench, lighting and providing safety to road users. But, these conditions are not fulfilled properly. At many places, heaps of earth (rubble, material, pipes etc) are seen lying on both sides of trenches for many days, creating problems for all pedestrians and vehicles. Hence, it is absolutely essential, to enforce the discipline by strict supervision and thorough penal action. The other inherent defect is in the backfilling process and materials used. These problem can be tackled by substituting Granular or other suitable material for entire backfill and compaction of pavement by vibratory rollers as per requirements (small size equipment/ compactors required) .

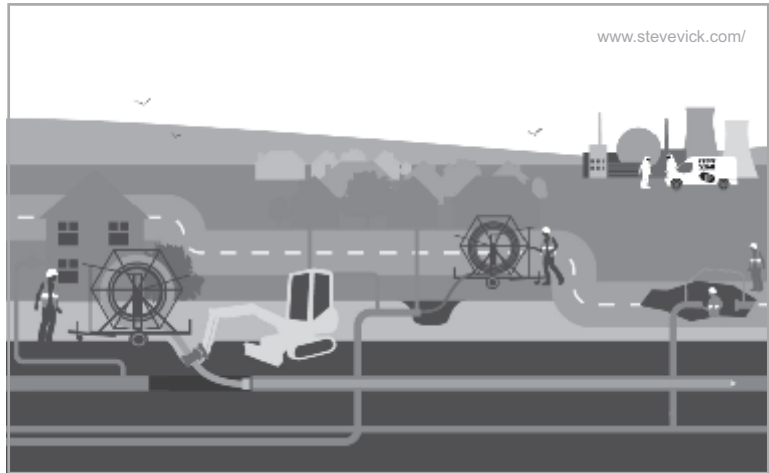
One more point of concern observed is the unauthorized excavations in the name of faults and emergencies by MSEDCL, Water Supply and Drainage&MNGL etc. The permission for excavation for trenching is not given between 30th April to 30th September, except under emergency conditions. The utility service provider is not provided any specific location for dumping of the excess material within PMC limit. Such practices need to be stopped by drastic deterrent actions through Pune Municipal Corporation vigilance squads, resorting to legal action, if required.

The permission for excavation for trenching is not given between 30th April to 30th September, except under emergency conditions. The utility service provider is not provided any specific location for dumping of the excess material within PMC limit. Such practices need to be stopped by drastic deterrent actions through Pune Municipal Corporation vigilance squads, resorting to legal action, if required.

Junior Engineer who supervises the work cannot give justice to the work due to other prior commitments. It is recommended, to appoint a separate Junior Engineer(at Zone level or Main department), exclusively for this work, during fair season and in monsoon season for pot holes and defect rectification of these works.

Use of Trench less Technology:

Use of Modern day techniques of trench less digging is an important solution for the problems faced for trench repairs. Techniques for Utility line installation, replacement, rehabilitation, inspection, location and leak detection are available in the market. These technologies require minimum digging at the ground surface.



Benefits of Trench less digging are as follows:

- 1.Safer for the public.
- 2.Minimal traffic impacts.
- 3.Minimal road repairs.
- 4.It reduces damages to valuable road surface.
- 5.It reduces the danger of improperly compacted excavation.
- 6.It saves resources and thus environmental degradation.
- 7.It reduces noise pollution.
- 8.It avoids annoyance to the public.

New Trench less Construction Methods

1. Impact Moiling.
2. Pipe ramming.
3. Auger Boring.
4. Pilot Tube.
5. Pipe Jacking.
6. Micro-tunneling.
7. Horizontal Directional Drilling.



Rajendra Raut
Chief Engineer (Road)
Pune Municipal Corporation

Guidelines for Granting of Permission To Utility and Municipal Agencies for Excavation and Reinstatement Thereafter

Part A-General

1. These Guidelines are applicable to all the utilities (internal and external), including all agencies/contractors of Pune Municipal Corporation, which will be excavating road surface for laying services.
2. For effective implementation of these Guidelines, there shall be proper co-ordination between the Ward Office, Main Road Department & utility/Municipal agency and reinstatement contractor (Ward Offices contractor or Main Road Dept contractor). Also there shall be proper coordination between the inter department to be done by lower level concerned Engineers.
3. Every Ward Office shall post at least One Engineer exclusively for Controlling all the activities of road excavation and its reinstatement and to assist him at least Two Mukadum (Mistry) level personnel shall be posted under him.
4. Pune Municipal Corporation Engineer should be on high vigil from time of granting the permission, up to the completion of the reinstatement work.
5. All proposals for digging have to be submitted by the Single window system operated by the Road department of Pune Municipal Corporation. From the single window system, the submitted proposals will be forwarded to the concerned Junior Engineer. The Utility personals should meet the concerned Junior Engineer for site visits and completion of various documents. After completion and requisite sanctions, the work order will be issued from the Single Window system only.
6. Utility agency/stake holder should submit weekly progress report to Pune Municipal Corporation (designated engineer) until completion of his trenching work, which shall be based on the detailed program submitted by him to PMC. The Engineer of Ward Offices & Main Road department shall submit weekly progress report with check list in the format to Executive Engineer as per **Annexure- 10 & 11**.
7. The Head Clerk of the Ward Office/Main Road Department shall maintain proper register in the format as **Annexure- 14**.
8. All reinstatement works shall be carried out through Main Road department and Ward Offices, except where a Municipal agency has incorporated reinstatement in its contracts to be carried out through its contractor.
9. To facilitate expeditious reinstatement, the tenders for reinstatement shall be approved before starting excavation.
10. The defect liability period for the reinstatement work carried out by Pune Municipal Corporation contractor shall be of one year from the date of completion of work.
11. For reinstatement works Security Deposit should be 5%, similar to other work. It will be recovered from the running bills of the contractor. Additional Quality performance guarantee of 5% shall be deducted from contractor's each payable bill and retained with the Municipal Corporation until completion of the defect liability period.
12. In case the trench gets damaged due to improper workmanship and/or material the Pune Municipal Corporation reinstatement contractor shall redo it by excavating the whole surface which he has undertaken earlier at his own cost.
13. On satisfactory performance and on completion of defect liability period the retention money will be returned with the sanction of the concerned Ex. Engineer/competent authority.

14. Point Nos.20,30,31,34,35,36,39,42 and other whichever applicable shall be followed by Pune Municipal Corporation while carrying out the reinstatement work and for violations penalties shall be as per Annexure- 12.
15. For expeditious completion of the work, The Vigilance Department may inspect the site and point out defect immediately, but they shall not instruct the stoppage of the work.
16. No other Municipal agency shall issue any circular in this regard without referring to this circular and the specific sanction of Pune Municipal Corporation.
17. Hon. General Body Resolution No.50 Dt. 20/5/2014 is applicable and binding to all utility agencies.
18. Circular of Government of Maharashtra DIT /File -03/336 Mantralaya, Mumbai Dt. 1/4/2006.is also applicable & binding to all utility Network agencies/ service providers.
19. Pune Municipal Corporation is developing a Mobile App for online Road excavation permissions which includes application to work order process, Complaint receiving and redressal which will work 24/7.

PartB-Laying/replacing of services

It is compulsory for all utilities service provider to use Trenchless technology like HDD, Pilot Tube etc. for laying of cables from October 2016. After June 2016, new permissions will be issued only for trenchless digging. Under special circumstances, if use of trenchless technology is not possible then other trenching methods will be allowed by the permission of Chief Engineer (Road) and it will be binding on service provider.

1. All the utility agency other than Pune Municipal Corporation dept. (water supply, drainage, electrical etc.) should submit their annual trenching programme with detailed proposal to road dept. before 31st August. Pune Municipal Corporation utility depts. shall submit their trenching programme before 30th Sep. with requisite budget provision. Small trenches for customer connectivity permission will be allowed throughout the year except monsoon.
2. Permission for laying, augmentation of the utility pipelines/cables shall be granted up to 30th Sept. Practically, excavation on site shall be started from 1st Oct. up to 31st March. For trenches, exclusively on footpath/ paver blocks, the permission shall be granted up to 30th April.
3. Normally, permission for digging roads between 1st April To 30th Sept. shall not be granted. Only permission for cable through duct / conduits may be granted in above period where excavation is not required. If any service provider or telecom operator is found to be illegally digging the road during this period he shall be penalized with appropriate legal action.
4. The utility provider proposing to undertake the trench shall make an application only in the prescribed format as per **Annexure-1**. Agency should submit two numbers of applications to Road department, Pune Municipal Corporation.
5. No incomplete application shall be entertained.
6. Route Map/Location sketch/trenching map should consist of proper GIS mapping with accurate measurement of length for trenching including the vertical cross section& strata wise bifurcation of length.
7. On receipt of application, form, concerned Engineer from Road dept. P.M.C. along with the representative engineer from Utility service provider and Engineer of Ward Office shall visit the site jointly and verify the details given in the application form in within 15 days period on receipt of application.

8. Inspection report form should be only in the format, as per **Annexure- 2** and if there is any variation, such variation shall be mentioned in the same format. This inspection report shall be signed jointly by Utility Engineer, Junior Engineer, and the Deputy Engineer.
9. Each application shall be duly scrutinized by the Pune Municipal Corporation's designated concerned engineer who may seek such additional information from the applicant service provider/ operator, as may be considered necessary for scrutiny of the application. While processing the said application, the concerned authority shall take into consideration, not only the existing infrastructure services and their safety and operations, but also future requirements of widening of the roads or augmentation of services.
10. Once the concerned Engineer is satisfied with the proposed alignment route/ site or such route as is determined and finalized mutually and also about the agreed time schedule in which the work is proposed to be executed, a Demand note shall be issued to the applicant. If Pune Municipal Corporation suggests another route instead of DIT sanctioned route, applicant should submit new DIT permission for that route only then, demand note will be issued .
11. If there is any variation in map/plan length, it shall be corrected by utility agency & it shall resubmit a corrected copy of plan to Road department. Then within 7 days period the proposal, with scrutiny report shall be forwarded to the Higher Authority.
12. Permission for taking up excavation in trench shall be granted by the Chief Engineer, Roads.
13. On receipt of sanction for trenching, immediately in within 3 days demand note along with estimate shall be sent to utility agency for payment of charges in the format as per **Annexure-7**. Payment shall be done within fifteen days after demand note is issued.
14. On receipt of payment from Utility agency on the basis of demand note, the Head Clerk shall make an entry of receipt of payment. Additionally, utility agency should submit Bank guarantee, an amount of 5% of demand note to the Head Clerk till completion report is submitted by the concerned Engineer.
15. On receipt of payments from utility agency, Road Dept. shall issue the permit (Work Order) within 3 days to the utility agency in the format as **Annexure- 8** with sanction plan/sketch, duly signed by concerned Junior Engineer and concerned Deputy Engineer. At the same time one copy of work order & plan shall be forwarded to the Ward Office and concern authorities. Also copy of work order shall be share on email to all members of steering committee.
On receipt of permission, Applicant/utility agency should publish the advertisement / public notice in local reputed newspaper regarding trenching work as per work order before start the work. Applicant must show / submit the public notice copy to the PMC before the start of work. Then after utility agency, shall immediately start the excavation and under no circumstances, shall they deviate from the permitted start and end point.
16. Time period for reinstatement by Ward office/ Road dept. shall be the date of completion of trench by utility agency plus maximum of 15 days.
17. In view of ensuring completion of trench within prescribed time limit, utility agency shall arrange all the resources well in advance before undertaking the excavation including obtaining permission from Traffic police etc. This permission concurrence and co-operation of Traffic Police. Copy of Permission must be kept with contractor on site supervisor to ensure no one creates stoppage by unnecessary objections. Concerned Junior Engineer of PMC should intimate to Traffic Police .
18. Permitted date of start and date of completion shall not be changed
19. Length more than 5% of trench shall not be increased under any circumstances.
20. Up to 5 % variation in length of trenching against permission, may be considered. Applicant shall do the work continuously but the reinstatement charges for excess length up to 5 % should be deposit in Pune Municipal Corporation within next 15 days period. If there is increase in the length by more than 5 %, then the Agency will be penalize 3 times the approved flat rate for entire extra length of work order. If for

unforeseen reasons the length increases above 5% and it is brought to the notice of PMC before the work is executed, then utility agency shall submit new proposal to PMC.

21. If Municipal Engineer is not agreeing with the date of start and completion as proposed by the utility agency in the original application, some other date for both start and completion, which shall be mutually agreed upon by the Municipal engineer & utility agency can be taken.
22. During the execution of related OFC works, the applicant shall ensure that no inconvenience is caused to the general public in the process of carrying out its operations. If found necessary, the competent authority may direct the applicant to undertake execution of its works during the off peak hours or during the night time.
23. Qualified engineer should be appointed by utility agency for all Trenching work & he should be report to Pune Municipal Corporation.
24. Time period for excavation & refilling shall be binding on utility agency. Black cotton soil/other soil shall not be permitted for refilling. The utility providers should themselves fill sand / granular material / non cohesive material in trenches for minimum of 30 cm above the utility top & leave a layer of tiles above it, so as to serve as a caution against any damage to the utilities.
25. Erection of boxes, pedestals, etc, above the ground shall not be permitted without specific sanction of Chief Engineer of road.
26. Erection of manhole, etc. below the ground shall not be permitted without specific sanction of Chief Engineer. In fact all manholes must be on level with road surface (not more than + 1 cm) not above or below.
27. The applicant shall undertake communication infrastructure works in a manner so as to cause least public inconvenience. He would be expected to suitably cordon off the area to ensure public safety and encouraged to execute the works during off-peak times so as to cause minimum inconvenience to the public traffic. The applicant shall restore the dug up area / sites and clear the area of an unused earth / debris, and dispose off such debris/earth/ rubble/excess material at the sites away from the work site.
28. The terms and conditions of guidelines issued/notified by the Department of Telecommunications, Government of India in respect of any conditions applicable to the infrastructure providers or licensees, as amended from time to time, shall be applicable and binding in all case.
29. Before starting the excavation, area shall be barricaded as stipulated in **Annexure- 9**.
30. Utility agency shall display the board at the start and end of trench and if the trench is more than 100 Meter, the board shall also be displayed in between and at the junction. If agency is failure to ensure that information display board on site, will attract penalty as per **Annexure-12**. Board must display information such as
 - (a) Name of Project
 - (b) Name of agency
 - (c) Date of starting
 - (d) Date of completion
 - (e) Name of site engineer
 - (f) Contact No. (Mobile)
 - (g) Name & contact no of PMC engineer
 - (h) With a sentence "Inconvenience caused is regretted."
31. Utility agency/applicant should submit trenching work completion letter to PMC within 7 working days after trenching work is completed. PMC engineer shall visit the site with engineer of utility Agency and if they have completed their work with lifting of debris etc. then he will issue the completion certificate within 7 days.

32. To avoid multiple excavations by different utilities, programme of laying of pipelines/cables shall be obtained from all utilities and synchronized well in advance in co-ordination with Traffic police.
33. Sufficient security guards and traffic wardens should be provided at the site of trenching.
34. Water entrances and chambers having opening shall be properly covered to avoid entry of excavated earth in it.
35. While carrying out excavation, it shall be seen that surface tiles/paver blocks, etc. are removed carefully and stacked properly .
36. Utility Site in-charge shall submit a weekly report to PMC engineer in the format as per **Annexure -10**.
37. At the appropriate locations, steel plates shall be provided at the trench for crossing of vehicular and pedestrian traffic.
38. Reflector signs shall be provided on the barricade all along the trench as **Annexure -9**.
39. If trench is more than 100 meter in length the utility shall only undertake 100 meter trench at a time, lay the pipeline/cable and clear the site for reinstatement and movement of traffic and only then further, excavation shall be undertaken. However in case of laying of power cable of long length in which unless complete length is excavated the cable cannot be laid, under such circumstances, 250 Meter may be allowed by separate sanction from Chief Engineer Road. If more length is required in such case, specific separate written sanction should be taken from Chief Engineer, Road dept. on merits.
40. It is likely that in the same trench, a number of agencies/ utilities may like to lay their pipeline /cables. Under such circumstances, the concerned utilities shall have proper coordination and cooperation with each other. But all utility agencies should take prior approval from Road Dept.
41. All the excavated earth shall be removed and transported within 24 hours.
42. On completion of reinstatement, the reconciliation statement by adding the amount of penalties for various lapses shall be prepared in the format as Annexure-13 and send to utility agency & Head Clerk of Road dept. either for refund of remaining deposit or recovery of charges as the case may be.
43. In case the utility fails to carry out excavation by the date of completion mentioned in the permit then with appropriate reason work permit may be given or the work order issued to the utility agency shall be cancelled by taking sanction of Chief Engineer Road dept. The copy should be sent Head Clerk Road dept. for refund /recovery of charges for a particular trench from the Utility Agency. The intimation to this effect should also be sent to Utility Agency.
44. In case, the utility undertakes excavation lesser than the permitted length, under the circumstances even though estimate is prepared as per this circular, work should be executed by Pune Municipal Corporation considering actual trench length as the permitted length and payment to PMC should be made accordingly. Similarly the balance charges recovered from the utility for reinstatement should also refunded there with.
45. In case the utility exceeds excavation in length, they shall be penalized as per Hon. General Body resolution No. 50 Dt. 20/5/2014. If deposit received from the utility is insufficient for extra work carried out, then letter shall be sent to the utility and no further permission shall be granted to the concerned utility unless they pay the penalty charges.
46. Wherever a new connection is required to be provided to the private party, such private party shall be treated as a utility under these Guidelines and all the procedures laid down shall be followed. It may not be possible to forecast the customer requirements in advance, Permission for connecting new customer shall be granted till 31st March only their trench lengths are up to 50 meter. That proposal shall be submitted separately.

47. As far as possible, excavations on carriage way shall not be permitted. However, if there is no alternative to lay a pipeline or cable then it may be provided as close as possible to the edge of carriage way leaving the water entrances. Utility agency shall be responsible for approval of line out for trenching from PMC engineer before start the work.
48. In case excavation is proposed to be taken across carriageway, condition of laying the pipelines/cable through duct and laying of additional 2 nos. of 300 mm Dia. N.P. Class II pipes or equivalent with encasing by M15 (1:2:4) concrete by utility agency shall be incorporated in the permit.
49. Laying of services under cement concrete and mastic finish roads shall not be permitted. Due to unavoidable circumstances, if a trench is proposed on the stretch under defect liability period and / or on c.c. pavement, prior approval shall be taken from Chief Engineer (Road).
50. In case, any shifting or change in alignment of the already laid optic fiber cable/other communication cables is necessitated due to widening of road/construction of flyovers or public building, the applicant shall be bound to do the same at his own cost within the period specified by the PMC. PMC should intimate the concerned utility agency at least 15 days in advance. If the applicant fails to comply with this condition to the satisfaction of the PMC authority the same shall be executed by the PMC at the risk and cost of the applicant. The charges so incurred on this account shall be recoverable from the applicant.
51. The applicant shall ensure safety and security of all underground installation/ utilities/ facilities and shall be solely responsible for compensation/ indemnification of concerned authority for damage caused/ claims or replacement sought for at the cost and risk of applicant to the concerned authority.
52. The extent of the digging trench should be strictly regulated so that cables are laid and trenches are filled up before the close of work.
53. The applicant shall make his own arrangement for crossing of cross drainage structure, rivers etc. below the bed. In case this is not feasible, the cables/ducts may be carried outside the railings/parapet wall and supported on brackets fixed to the outside of the bridge or super structure. The fixing and supporting arrangement with all details shall be got approved in advance from the concerned authority granting permission. Fixing and supporting arrangement to be done applicant.
54. In case of any damage to the essential services i.e. Water supply, sewerage system and Tele communication lines, electricity supply etc, it will be the responsibility of the Digging Agency to get the services restored to their original and satisfactory condition at its own cost.
55. The applicant shall have to provide barricading, safety warning lighting and other necessary caution boards, warning lights, reflectors etc while executing the works.
56. If any traffic diversion works are found necessary during the working period, such diversion shall be provided by the applicant at his cost.
57. The PMC authority will be competent to effect any modification/alterations in the site plan/route, if necessary, in the interest of public safety.
58. If for any reason, it is not possible to carry out the actual work, as per the alignment given in the trenching work order, then the applicant shall not change the alignment without taking prior approval from Chief Engineer.
59. Applicant (Telecommunications service provider) should provide free network connection as per mentioned in DIT circular / Agreement as soon as the network is commissioned. It should be noted that 5% bank guarantee will be released after this provision only.
60. Micro trenching will be allowed if proposal given by Agency.

61. Applicant should not increase no. of ducts as approval from DIT. Applicant shall not transfer/lease/commercial use / sale to the other agency.
62. Use of Micro-trenching will be encouraged and preferred for cable laying activities.
63. For excavations at important Junctions and Cross cuts for Road with ROW 12 m and above, it is mandatory on the concerned OFC Utility Agency/MNGL/MSEDCL,etc to use HDD technology. In city core Peth areas or in congested areas, Use of HDD technology has to be done by the OFC utility agency/ MNGL / MSEDCL, etc. Any change in the above will be done only with the permission of Chief Engineer (Roads).
64. Water supply Department and Drainage departments should use trench less technology or pipe bursting methods especially for core city areas.



Part C-Reinstatement:

1. If the excavation is proposed on a footpath/ passages, irrespective of whatever may be original surface finish, flat rate with paver block surface shall be considered for preparing Demand Note, estimate for reinstatement and also work shall be done only with the paver blocks.
2. If excavation is proposed on the carriage way, the flat rate of original surface finish shall be considered for preparing demand note, estimate and reinstatement.
3. Work order to reinstatement contractor shall be given in the format as ,approved by the Pune Municipal Corporation.
4. Wherever required, raising of kerb stone shall be considered while preparing the estimate.
5. While preparing the estimate, at least 10% of length of the trench shall be considered for replacement of kerb stone.
6. The backfilling of trench shall be done simultaneously by Ward office/concerned dept. only with graded granular material. While preparing an estimate, sufficient quantity of graded granular material with watering and ramming shall be considered.
7. In case utility/ reinstatement contractor/ Ward office fails to start the work by the date of completion mentioned in the permit and seeking revalidation of the permit it maydone with the specific sanction of the Chief Engineer road.
8. If it is found and pleaded by the utility/ reinstatement contractor/ Ward office that the circumstances due to which their activated are delayed were beyond their control, under the circumstances the penalty may be waived off by only the Chief Engineer Road by hearing and passing order to the effect.
9. At the time of reinstatement, the width of trench should be such that compaction can be done by small vibratory roller and hand compaction should not be permitted in any case.
10. On cable work and at the crosscuts of the road , trench filling work shall be done using CLSM material / M-40 concrete.
11. One Junior Engineer should be made exclusively available for trench work for each ward to closely super wise the work and ensure adherence to the guidelines and conditions.



Part D-Fault Repairs and Emergency Excavation Digging Permission

1. As far as possible, all utilities including Municipal agencies shall seek prior permission before undertaking excavation for fault repair. However, in case of severe emergencies, applicant should intimate to PMC engineer on mail / sms and written application should be submitted to Ward office / Road Department in first 2 hours of next working day.
2. Executive Engineer/ Assistant Municipal Commissioner shall grant permission for repairing fault within 3 days after receiving the application. Also all utility agencies shall seek prior permission for replace the damaged portion of cables between to MH/HH without any digging work from Executive Engineer.
3. It is often observed that even though there are alternate arrangements established, the utility undertakes excavation under the guise of fault repair. This should be absolutely barred and it will penalized as per **Annexure- 12**.
4. For the purpose of reinstatement of faults, sufficient amount shall be deposited by the utilities as decided by Chief Engineer Road.
5. Between 16th April to 30thSeptember, for the fault repair, sanction of Add.. Municipal Commissioner shall be obtained as far as possible before undertaking excavation and in no case, excavation shall be taken more than 5 Meter in length and 1 meter width without appropriate sanction irrespective of whatever emergency. However, in case of fault in the tension power cable (more than 22 KV) the trench up to length of 20 meter may be considered on merits with due sanction by the Chief Engineer of Road Department.
6. Utility shall start excavation, repair the fault, remove excavated earth, etc. within maximum of 48 hours.
7. If excavation is undertaken by various Municipal agencies like Water Works, Sewerage Operation, S.W.D for taking pressure, repairing pipe lines, etc., all conditions laid down shall be followed.
8. Concerned Municipal agencies shall ensure that they follow proper procedure for reinstatement tender before starting the excavation or provide fund (give locking) to Ward office/ Road department to facilitate the reinstatement of trenches. Also concerned Municipal agencies shall ensure sufficient financial provisions in their budget.
9. The instructions mentioned in Part B Laying / replacing services shall also be followed for this part as well.
10. Reinstatement of trenches taken for fault repairs shall be done through Ward Office or through concerned agency of Road Dept, if there is provision in the contract.
11. For effective reinstatement of excavation made for fault repair, Municipal Engineer shall consolidate utility-wise all excavations taken for fault repairs in the preceding week and on every Tuesday, he shall prepare proper plan and issue the order to contractor on the basis of tender.
12. Reinstatement contractor shall carry out reinstatement during dry spell within next Two days. This process shall be followed every week.
13. For preparing an estimate, width of reinstatement on footpath/ passages shall be same as that mentioned for laying of new services. Similarly surface finish shall only be with paver blocks. As regards length, it shall be an additional one meter on either side of the actual trench.
14. If repair of fault is on carriageway, estimate shall be prepared for 2.4 Meter width with original surface finish.
15. In no case, a trench taken for fault repair by the utility shall remain to be reinstated for more than Seven days.
16. Around the junction and ADP Boxes, signals, etc, large number of cables are laid thereby frequent excavation is observed may be due to granting of additional connections, faults, etc. which not only cause inconvenience to the Municipal Corporation and citizens but also to the utility. Therefore, to avoid frequent excavation the Utility Agency shall be insisted to construct a duct around such boxes and locations.

Part E- Street Lighting

In Case of laying of cable or repairing of faults for street light & signal works, the procedure for issuance of permit, preparation of estimate and reinstatement shall be as per Guidelines, except non issuance of Demand Note, as the reinstatement is to be carried out at the cost of Municipal Corporation.

1. The fund required for that reinstatement should be provided by concerned street light Dept./Traffic dept. to ward office/ Road dept. Therefore, for noting and booking of liability, the estimate prepared (excluding water supply, sewerage & supervision charges) by Ward office shall be sent to Executive Engineer (street light / Traffic) who shall note the liability within two days and after locking the amount send it to the concerned Ward Office / Road dept.
2. Only after the arrival of estimate with locking to Ward office, permit to the utility and work order to the concerned utility agency will be issued.
3. In this case, since we are not taking deposit from the utility, penalties shall be recovered from the same Utility Agency undertaking trenching activity elsewhere, for providing their services.

Part F-For Municipal Utilities

The procedures mentioned in this guidelines shall also be followed for laying of pipelines by the Municipal Utilities except for the following:

- 1) Since Municipal water and sewerage lines are required to be laid at the deeper level, time period required for completion may be more. However, reinstatement shall be as per revised methodology/rates by incorporating provision in the tender. For deeper trenches, granular filling below road crust may be restricted to 10 layers (each of 150 mm thickness) over well compacted back filled earth/materials.
- 2) For departmental works taken up by Municipal utility Dept. the fund required for reinstatement should be provided by Municipal utility Dept. Therefore, for noting and booking of liability, the estimate prepared by Ward office shall be sent to Municipal utility Dept. who shall note the liability within 2 days and send it to the concerned Ward office with locking.
- 3) For Failure to comply with the stipulated time limit, the penalties shall be recovered as per **Annexure-12** from the contractor responsible for laying the lines. Similarly, penalties are also required to be levied for reinstatement of trenches from the concerned contractor.
- 4) The agency who has been permitted to undertake excavation must submit Weekly report in the format given in **Annexure 10** to the PMC engineer.
- 5) The Condition Nos.10,11,12 and 13 of Part A- General, of this circular shall also apply to the reinstatement contractor of the Municipal agency.

Part G- Methods of reinstatement of trenches

- 1) On Laying of services it shall be properly covered and secured with tiles and/ or granular material by the utility. The thickness of such covering shall be sufficient so that while compaction of granular material and reinstatement of trenches the services laid should not get damaged.
- 2) The reinstatement contractor shall fill the trench with the granular layers. Each layer of granular material shall not be more than 150 mm in thick. Sufficient watering and ramming of each layer shall be done to avoid settlement of trench in future.

- 3) Over the layers of granular material, further reinstatement shall be done as per **Annexure-4,5 & 6** as directed.
- 4) All reinstatement work including Asphaltting shall be completed prior to 10th May.

Part H- Monitoring/reporting status of trenching activity

1. It is necessary to have centralized co-ordination and Monitoring mechanism so as to make administration / Public / traffic police aware of such activity & its status from time to time.
2. All concerned utility Agency are therefore required to invariably forward utility wise/ ward wise details of trenching activity in the format given in Annexure-16. Report for every month must reach the office of Chief Engineer Road by Date 5th of the Every month.



Annexure 1 to 17

Annexure -1

APPLICATION FORM FOR TRENCHING
(On the letter head of Utility /Municipal Agency)

Date :-

To,
Chief Engineer (Road)
Pune Municipal Corporation

Sub :- Excavation of Trench on-----

Dear Sir,
We would like to undertake excavation in trench for laying of our service utility. The details of proposed trench is as follows.

1. Name of the road -----
2. Exact location of starting point
- A) Name of Place
- B) GIS coordinates of the Place -----
3. Exact location of end point
- A) Name of Place
- B) GIS coordinates of the Place -----
4. Width of Trench -----
5. Total Length of Trench -----
6. Length of trench on footpath/passages -----
7. Length of Trench on carriageway -----
8. Existing surface of carriageway -----

No.	Road Surface	Length in Mtr.
1	Asphalt Road	
2	Mastic Asphalt	
3	Cement Concrete	
4	Crossing Duct (PVC/RCC Pipe/RCC Duct)	
5	Footpath	
6	others	
7	Total	
8	Details of services on proposed road	
9	Proposed required period for work	
10	Name of the site engineer	Mobile No Office Tel/ No.
	Whether scaled plan showing all above details attached.	
12	Whether reconnaissance Survey of location for determining the feasibility of trench is to be taken is carried out or not.	Yes/No
13	Whether DIT permission as per route name asked is received and copy of same attached with the application(for OFC applications only).	Yes/No
14	Whether MSDCEL permission as per route name asked is received and copy of same attached with the application(for MSDCEL applications only).	Yes/No

Please examine our above request and kindly permit us to take excavation to lay our service. Kindly, also inform the reinstatement charges to be paid by us.

We also agree to abide by conditions enforced laid down by the Municipal Corporation in this regard from time to time.

Thanking You,

Yours faithfully,
(Name of .co. Name, designation, Contact No.)

Enclosure :- 1) DIT Permission
2) GIS mapping soft copy.
3) GIS mapping hard copy.
4) Water Supply / drainage / electrical / MSEDCL permission

Annexure -2

PUNE MUNICIPAL CORPORATION
JOINT INSPECTION REPORT

Sub:- Excavation of Trench on -----

Ref:- Contractor letter to P.M.C. dated :-

I, along with Shri ----- Engineer of Utility of
M/S ----- and Junior engineer of the area -----
-----inspected the site of ----- proposed trench work at above
location on -----at-----A.M./P.M. and found
that the details mentioned in
application form are/are not in order and following points are observed.

1. Name of the road -----
2. Exact location of starting point
- A) Name of Place
- B) GIS coordinates of the Place -----
3. Exact location of end point
- A) Name of Place
- B) GIS coordinates of the Place -----
4. Width of Trench -----
5. Total Length of Trench -----
6. Length of trench on footpath/passages-----
7. Length of Trench on carriageway -----
8. Existing surface of carriageway -----

No.	Road Surface	Length in Mtr.
1	Asphalt Road	
2	Mastic Asphalt	
3	Cement Concrete	
4	Crossing Duct (PVC/RCC Pipe/RCC Duct)	
5	Footpath	
6	others	
7	Total	

Details of services on proposed road _____

Approximate Quantity of excavated earth _____

Name of the site engineer _____

Signature of Engineer of Utility
Engineer

Signature of Junior

Road dept.

M/S

The above report may please be pursued. The Utility M/s -----
----- has requested for taking excavation in trench on-----
The site was inspected jointly.

If agreed the excavation of ----- meters length on footpath/carriage way will be permitted. The date of start is ----- and reinstatement charges of trench digging amount to
Rs-----is prepared and reinstatement will be carried out through Road Dept.

Ms-----

Submitted please.

Deputy Engineer (Road dept.)
Pune Municipal Corporation
Agree/Disagree
Executive Engineer

Annexure -3

PUNE MUNICIPAL CORPORATION SCHEDULE OF FLAT RATE FOR EXCAVATION

As per Hon. Standing committee resolution No. 177 dt. 13/5/2014

No.	Type of Road	Rate in Rs. Per Rmt.
1	PVC Pipe	5950/-
2	RCC Pipe	5950/-
3	Asphalt Road Reinstatement charges	5547/-
4	Footpath Reinstatement charges	5488/-
5	By Use of HDD Machine	3000/-

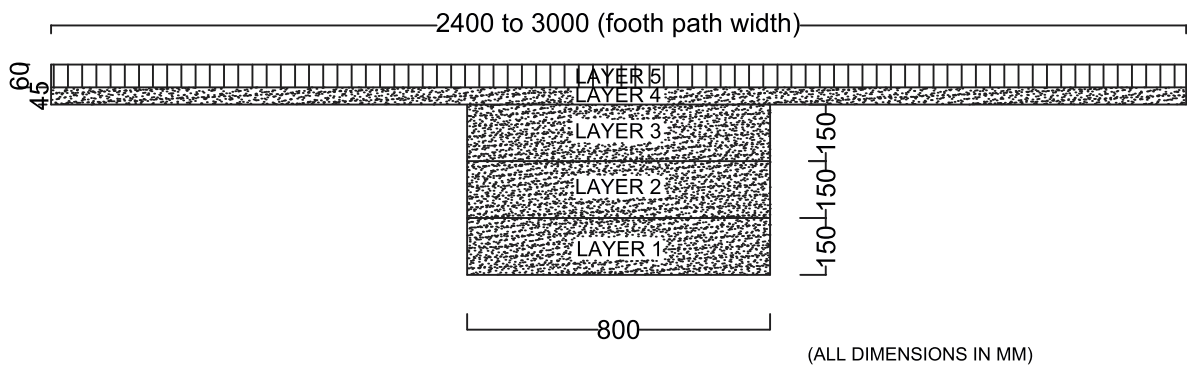
Excavation Pit reinstatement charge -6000/- per sq.mt.{at the chamber position)

Note :-

- 1) The above flat rates are calculated on the basis of PMC DSR 2014-15
- 2) The flat rates are approved by Hon. Standing Committee resolution No. 177 dt. 13/5/2014.
- 3) The flat rates are subject to revision in accordance with revision of PMC DSR.

Annexure -4

TYPICAL CROSS SECTION OF TRENCH FOR
REINSTATEMENT ON FOOTHPATH/PASSAGES



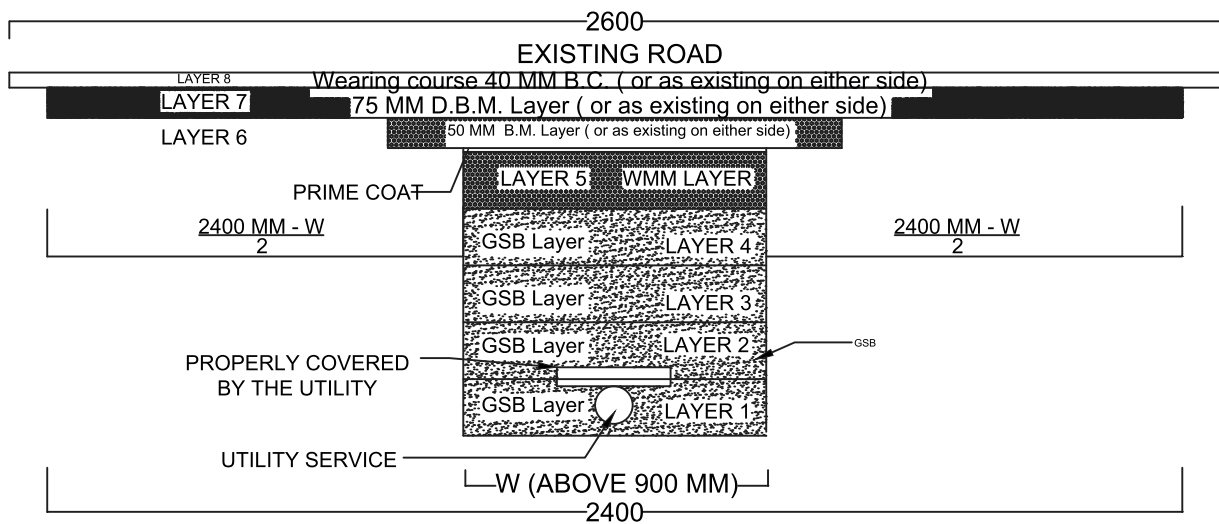
Layer wise details for item/materials to be
used & its compaction

Layer No.	PMC Item DSR code no.	Discription of Material	Remarks
Layer 1	Road part 2-8-1 88815	Granular Sub base Material	Sufficient watering & Ramming by Utility
Layer 2 & 3	Road part 2-8-1 88815	Granular Sub base Material	Sufficient watering & Ramming by Reinstatement contractor
Layer 4 + 5	Road part 6-2-1 88941	Grit/Crushed sand/ River sand cushioning & C. C. paver Block	-----

- NOTE : 1) The thickness and No. of Layers Granular Sub base Material filling will depend on proposed depth of excavation as per application from utility Annexure - 1
- 2) The maximum thickness of each Layer of Granular Sub base Material shall not be exceed 150 mm.

Annexure -5

TYPICAL CROSS SECTION OF TRENCH FOR REINSTATEMENT ON ASPHALT ROAD

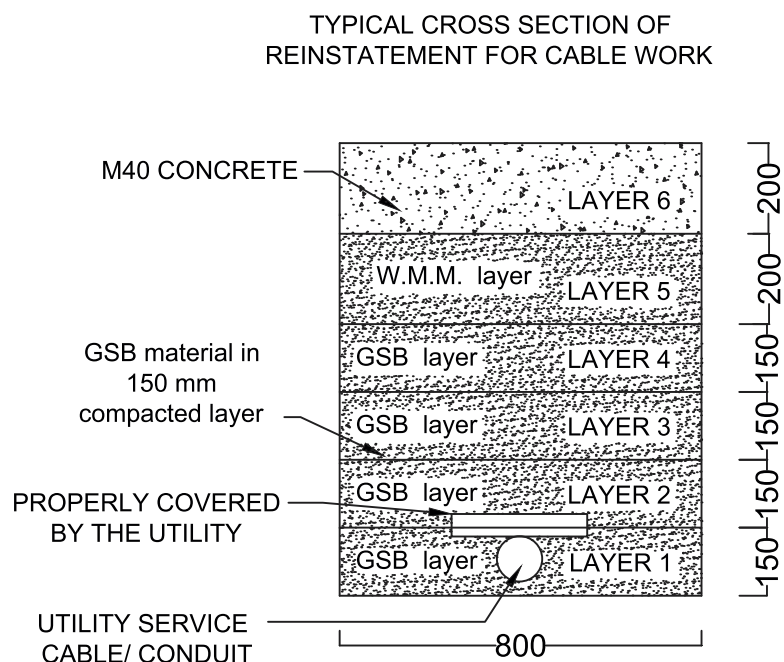


Layer wise details for item/materials to be used & its compaction (ALL DIMENSIONS IN MM)
NOT TO SCALE

Layer No.	PMC ITEM DSR code no.	Discription of Material	Remarks
Layer 1 & 2	Road part 2-8-1 88815	Granular Sub base Material	Sufficient watering & ramming by utility
Layer 3 & 4	Road part 2-8-1 88815	Granular Sub base Material	Sufficient watering & ramming by reinstatement contractor
Layer 5	Road part 2-10-1 88817	W.M.M. material	Comaction by 10 Tonnes vibratory Roller by contractor
Layer 6	Road part 4-4-2 88835	Bituminous Macadam	laying by Paver finisher and Compaction by 10 Tonnes Vibratory Roller
Layer 7	Road part 4-3-2 88833	Dense Bituminous Macadam	
Layer 8	Road part 4-5-2 88837	40 mm thick Bituminous Concrete	

- NOTE : 1) The thickness and No. of Layers Granular Sub base Material filling will depend on proposed depth of excavation as per application from utility Annexure - 1
- 2) The maximum thickness of each Layer of Granular Sub base Material shall not be exceed 150 mm.
- 3) Vibratory roller shall be used for compaction of each layer of road crust.
- 4) For deeper trenches for internal road of P.M.C. granular filling sub base below road crust may be restricted to maximum 10 layers (each of 150 mm. thickness) over well compacted backfilled earth/materials.

Annexure -6



Layer wise details for item/materials to be used & its compaction

Layer No.	PMC Item DSR code no.	Discription of Material	Remarks
Layer 1 & 2	Road part 2-8-1 88815	Granular Sub base Material	Sufficient watering & Ramming by Utility
Layer 3 & 4	Road part 2-8-1 88815	Granular Sub base Material	Sufficient watering & Ramming by Reinstatement contractor
Layer 5	Road part 2-10-1 88817	W.M.M. Material	Sufficient watering & Ramming by Reinstatement contractor
Layer 6 M40 CONCRETE	Road part 5-15-3 89147	M40 CONCRETE	-

NOTE : 1) The thickness and No. of Layers Granular Sub base Material filling will depend on proposed depth of excavation as per application from utility Annexure - 1

2) The maximum thickness of each Layer of Granular Sub base Material shall not be exceed 150 mm.

Annexure -7

PUNE MUNICIPAL CORPORATION DEMAND NOTE

No. _____

dt. _____

To,

Sub. : - Trench on _____

Ref. : - 1) Circular No.

2) Your letter under no. _____

Dt. _____

With reference to your above referred letter the site was visited and your request to take trench has been accepted in principle, subject to payment of Rs. _____ as follows.

No.	Particulars	Length in Mtr	Rate in Rs.	Per	Amount in Rs.
1	Asphalt Road				
2	Mastic Asphalt				
3	Cement Concrete				
4	Crossing Duct/ pipe				
5	Footpath				
6	Others				
	Total				

You shall make payment to the office of Chief Engineer, Road office, Pune Municipal Corporation, by separate D.D. of Rs. (only nationalized bank) towards reinstatement charges and deposit it in favour of Municipal Commissioner, Pune Municipal Corporation.

In addition, you shall also submit a Bank guarantee of 5% of total amount of Rs. (only nationalized bank). On payment you shall present the copy of the receipt and original Bank guarantee to PMC office, so that the necessary permission can be granted. You shall make payment within Fifteen days from receipt of this demand note failing which this demand note will become invalid.

Further also note that payment of reinstatement charges to Pune Municipal Corporation means does not guarantee the grant of permission.

Yours faithfully,
Executive Engineer (Road Dept.)
Pune Municipal Corporation

Copy to :Head Clerk

You are hereby requested to take note of payment of reinstatement charges and Bank guarantee. Also, prepare the work order, take the signature of Executive Engineer and issue it to the Utility Agency. One copy of work order should also be sent to the Ward Office.

H.C.(Rev.) _____ Ward Office.
Pune Municipal Corporation.

Annexure -8

FORMAT OF WORK ORDER TO BE ISSUED TO THE UTILITY AGENCY

Chief Engineer (Road)
Pune Municipal
Corporation
Out word No –
Date-

M/s

Sub. : - Trench on

Ref. : - 1) Circular No.

2) Your application no._____ dt._____.

3) Demand note under no._____.

4) Charges of reinstatement amount Rs.-----Challan No. Date

Sir,

With reference to your request, Excvation Activity for the following trench is permitted :-

1. Starting Point
 - A) Name of Place
 - B) GIS coordinates of the Place
2. End Point
 - A) Name of Place
 - B) GIS coordinates of the Place
3. Length of Trench
4. Date of Start
5. Date of Completion

Subject to the following conditions:-

1. Pune Municipal Corporation Guidelines for Trenching Activity shall be applicable/ binding on the applicant/ service providers/ utility agency and is available on website.
2. On receipt of permission, you shall immediately start the excavation.
3. Above mentioned date of start and completion, length of Trench, start and end point of Trench shall not be changed under any circumstances.
4. If Municipal Engineer is not agreeing with the date of start and completion proposed by the utility agency and if he is suggesting any other date for both start and completion, this shall be communicated to and agreed upon by the utility agency.
5. To avoid multiple excavations by different utilities, program of laying of pipelines/cables shall be obtained from all utilities and synchronized well in advance in co-ordination with traffic police.
6. It is likely that in the same trench, a number of agencies/ utilities may like to lay their pipeline /cables. Under such circumstances, the concerned utilities shall have proper coordination and cooperation with each other. But all utility agencies have to take prior approval from Road Dept.
7. As far as possible, excavations on carriage way shall not be permitted. However, if there is no alternative to lay a pipeline or cable then it may be provided as close as possible to the edge of carriage way leaving the water entrances. Utility agency shall be responsible for approval of line out for trenching from PMC Engineer before start the work.
8. Before starting the excavation, area shall be barricaded as per annexure-9 all along the excavation of site trench and warning tapes shall be placed on both side of trench so that no accidents should happen.
9. Utility shall display the board at the start and end of Trench and if the Trench is more than 100 Meter it shall also be displayed in between and at the junction. Board must display information such as
 - (A) Name of Project
 - (B) Name of agency.
 - (C) Date of starting.
 - (D) Date of completion.
 - (E) Name of site engineer.
 - (F) Contact no.(mobile)
 - (G) with a note that "Inconvenience caused is regretted".
10. Sufficient security guards and traffic wardens should be provided

11. Water entrances and chambers having opening shall be properly covered to avoid entry of excavated earth in it.
12. Your site in charge shall submit the weekly report to P.M.C. Engineer in a format as per Annexure - 10. Of Guideline for Trenching activity for Pune City.
13. Wherever required steel plate shall be provided all along the Trench.
14. Reflector signs & night lighting shall be provided all along the Trench.
15. If Trench is more than 100 Meter in length, you shall only undertake 100 Meter Trench at one time, lay your services and clear the site for reinstatement and open road for movement of traffic and then only, further excavation shall be undertaken.
16. It is likely that in the same Trench, number of agencies/ utilities may lay their services. Under such circumstances, you shall have absolute co-ordination and co-operation with each other.
17. All the excavated earth shall be transported .Immediately away from the site.
18. Time period for excavation & refilling shall be binding on utility agency. Black cotton soil/other soil shall not be permitted for refilling .Only granular sub base material shall be allowed for refilling.
19. Erection of boxes, pedestals, etc, above the ground shall not be permitted without specific sanction of Chief Engineer of road.
20. The Applicant shall undertake communication infrastructure works in a manner so as to cause least public inconvenience. He would be expected to suitably cordon off the area to ensure public safety and encouraged to execute the works during off-peak times so as to cause minimum inconvenience to the public traffic. The Applicant shall restore the dug up area/ sites and clear the area of an unused earth/debris, and dispose off such debris/earth at the sites away from the work site.
21. In case of taking Trench across the carriageway at least 2 number of 300 mm. Dia. N.P. II class Hume pipe or equivalent with encasing of 1:2:4 concrete shall be laid for any further service. Also you shall lay your services through duct.
22. Simultaneously, work order for reinstatement is issued to Ward Contractor and he will take up the work on the date of completion of your work mentioned in this permit or in phases as per clause no.14 above.
23. The Qualified Engineer should be available at all the time during progress of work on site and he should report to PMC. Also one copy of work order and approved drawing and maps should be on site with responsible person as and when required by PMC.
24. On receipt of work order, you shall immediately contact all other utility intimating that you will be taking Trench along the particular stretch and also arrange all the resources required by you for expeditious completion of your work.
25. If there is road crossing or any work that obstruct safe pedestrian movement or traffic movement you will have to take NOC from traffic police dept. before starting of work.

26. By virtue of any circumstances, you are unable to undertake excavation as per this work order, you shall immediately communicate to the undersign. Under the circumstances you shall note that same permit will not be granted in future.
27. You shall not to take Trench in Cement Concrete / Mastic finish surfaces.
28. Wherever the existing finish is with paver block or lacquered colour paver block you shall not break / damage these blocks while excavation & shall see that almost 100% of the blocks are removed intact & handed over to Municipal Engineer/Ward office Engineer.
29. Excavation in asphalt should be done by using breaker only & use of JCB for excavation in asphalt should be avoided as far as possible.
30. Utility Agency/applicant should submit trenching work completion letter to PMC within 7 working days after trenching work is completed. PMC Engineer shall visit the site with Engineer of utility Agency and if they have completed their work with lifting of debris etc. then he will issue the completion certificate.
31. In case any shifting or change in alignment of the already laid optic fiber cable/ other communication cables is necessitated due to widening of road/construction of flyovers or public building, the Applicant shall be bound to do the same at his own cost within the period specified by the PMC. If the applicant fails to Comply with this condition to the satisfaction of the PMC Authority the same shall be got executed by the PMC at the risk and cost of the Applicant. The charges so incurred on this account shall be recoverable from the applicant.
32. The applicant shall ensure safety and security of all underground installation/ utilities/ facilities and shall be solely responsible for compensation/ indemnification of concerned authority for damage caused/ claims or replacement sought for at the cost and risk of applicant to the concerned authority.
33. In case of any damage to the essential services i.e. Water supply, sewerage system and telecommunication lines, electricity supply etc, it will be the responsibility of the company to get the services restored to their original and satisfactory condition at its own cost.
34. If any traffic diversion works are found necessary during the working period, such diversion shall be provided by the Applicant at his cost.
35. As far as possible existing duct/RCC pipe /PVC pipe shall be utilize for passing services as per instruction of concern Engineer. Existing ducts are the property of PMC. So applicant shall not do any work (cable passing, fault repair etc.)without approval of PMC
36. Before excavation machine detector shall be used for searching alignment of existing underground pipeline/ cable etc.
37. Applicant/utility agency should publish the advertisement / public notice in local reputed newspaper regarding trenching work as per work order before start the work. Applicant must show / submit the public notice copy to the PMC before start the work.
38. Applicant should not increase no. of ducts as approval from DIT. Applicant shall not transfer / lease / commercial use / sale to the other agency.
39. Trenching should not be allow under the Railway bridge .

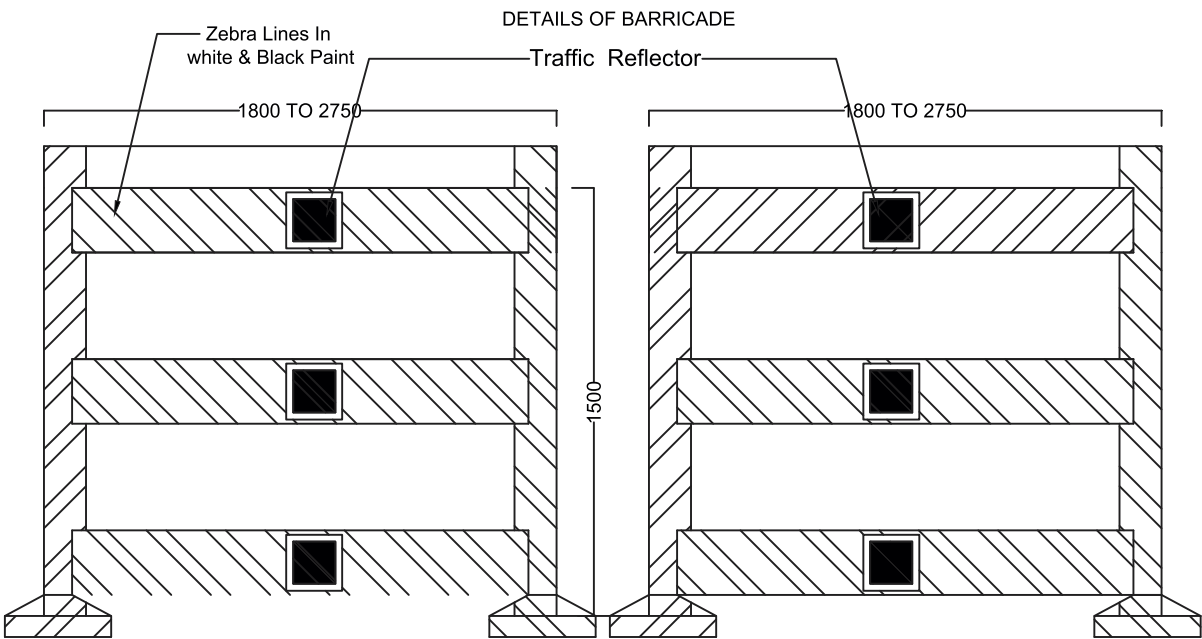
40. Hon. General Body Resolution No.50 Dt.20/5/2014 is applicable and binding to all utility agencies.

41. Circular of Government of Maharashtra DIT /File -03/336 Mantralaya, Mumbai Dt. 1/4/2006.is also applicable & binding to all utility Network agencies/ service providers.

Executive Engineer
(Road)
Pune Municipal Corporation

Copy to: 1) Deputy Engineer.
 2) Concerned ward office
 3) Head Clerk

Annexure -9



Barricade can be made from Steel or Wood

NOT TO SCALE

DETAILS OF DISPLAY BOARD

600	900	
	Name of Agency	_____
	Date of Starting	_____
	Date of Completion	_____
	Name of Site Engineet	_____
	Contact No. (Mobile)	_____
INCONVENIENCE CAUSED IS REGRETTED		

Annexure -10

(On the letter head of Utility)
WEEKLY PROGRESS REPORT OF UTILITY

To,
Deputy Engineer

Sub:- Trench on -----

Ref :- Work Order No. _____ dated _____
Sir,
1. Work Start Date :-

2. Date of Progress Report :-

3. As per work order work Complete Date :-

The progress of above work is as follows :-

No	Particular	Proposed as per plan	Actual Progress
1	Laying of cable		
2	Transporting of excavated earth		
3	Damage to Municipal Utility		
4	Damages to other Utility		
5			

Date:- _____ Engineer of Utility.

Observations of Junior Engineer

1.	Delayed in starting	Yes/No
2.	Delayed in completion	Yes/No
3.	Engineer of Utility is available	Yes/No
4.	Barricading fixed	Yes/No
5.	Refectory signage provided	Yes/No
6.	Name Board displayed	Yes/No
7.	Warden appointed	Yes/No
8.	M.S. Plate provided	Yes/No
9.	Earth removed in time	Yes/No
10.	Excavated earth removed	Yes/No
11.	Water entrances covered	Yes/No
12.	Water entrances cleaned	Yes/No
13.	Municipal Utility damaged	Yes/No
14.	Other Utility damaged	Yes/No
15.	Damages to private property	Yes/No
16.	Length increased	Yes/No
17.	Alignment changed	Yes/No
18.	Starting Point changed	Yes/No
19.	End point change	Yes/No
20.	Missing cover provided	Yes/No
21.	Night lighting provided	Yes/No

Date :-

Junior Engineer
Road Dept.

Annexure -11

WEEKLY PROGRESS REPORT OF REINSTATEMENT CONTRACTOR

Sub:- Trench on-----

Ref:- Work order No. Date:-

Sir,

- 1. Work order No.
- 2. Name of Agency.
- 3. Date of Progress Report.

The progress of above work is as follows:-

No.	Particular	Proposed as per	Actual Progress
1	Backfilling in trenches with approved material (in meters)		
2	Metelling/GSB (in meters.)		
3	Bituminous Work(in meters.)		
4	Concreting (in meters.)		
5	CLSM (in meters.)		
6	Fixing C.C. paver Block		

Observations of Junior Engineer

1.	Delayed in starting	Yes/No
2.	Delayed in completion	Yes/No
3.	Engineer of Utility is available	Yes/No
4.	Barricading fixed	Yes/No
5.	Refectory signage provided	Yes/No
6.	Name Board displayed	Yes/No
7.	Warden appointed	Yes/No
8.	M.S. Plate provided	Yes/No
9.	Earth removed in time	Yes/No
10.	Excavated earth removed	Yes/No
11.	Water entrances covered	Yes/No
12.	Water entrances cleaned	Yes/No
13.	Municipal Utility damaged	Yes/No
14.	Other Utility damaged	Yes/No
15.	Damages to private property	Yes/No
16.	Length increased	Yes/No
17.	Alignment changed	Yes/No
18.	Starting Point changed	Yes/No
19.	End point change	Yes/No
20.	Missing cover provided	Yes/No
21.	Night lighting provided	Yes/No

Date :-

Junior Engineer
Road Dept.

Annexure -12

PUNE MUNICIPAL CORPORATION SCHEDULE OF PENALTIES

Schedule of Penalties to be recovered from utility/ Reinstatement Contractor for various lapses observed during the excavation and reinstatement of trenches.

(A) For Utility and reinstatement contractor.

No.:	Particular	Penalty in Rs.	Per
1.	Non fixing of barricading at site	1000	Meter
2.	Non displaying information board at site @ 100-meter distance	500	Meter
3.	Not Providing signage boards and reflectory signage's at the site	500	Meter
4.	Non appointing of warden/assistance at crowded traffic road per site	200	Day
5.	After cable laying / service line work complete, non-removal of excavated balance material /soil from site in time per site	1000	Day
6.	Not cleaning storm water /drainage line Chamber if excavated soil fall in chamber	1000	Per chamber
7.	Damage to Municipal utility/Private property/ Other utility	Actual Cost	
8.	Increase in trenching length against sanctioned work order – (More than 5%of approval)	Three times of approved flat road per Meter	

(B) For reinstatement Contractor.

No.:	Particular	Penalty in Rs.	Per
1.	Not rolling and compacting as per specification.	1000	Meter
2.	Settlement of trench in defect liability period if not doing rectification	2000	Meter
3.	Non attending of damaged trench within 7 days during defect liability period from the date of receipt of the letter	1000	Sq. meter.
4.	Attending damaged trench through other Agency after 7 days during defect liability period	Actual Cost at his risk and cost.	

Note :-Wherever unit is mentioned as meter it shall be taken with reference to length of trench.

Annexure -13

PUNE MUNICIPAL CORPORATION
RECONCILIATION LETTER UTILITY AGENCY

To,

Sub. : - Trench on _____

Ref. : - 1) Circular No.

2) This office Work Order no. _____ dt. _____

Sir,

With reference to above Work Order, you have excavated a trench of----- meters length as against the sanctionedmeters length. Thereby you have exceeded/ reduced the length of trench by _____ meters.

The Reinstatement Charges to be recovered are as follows.

No.	Particular	Flat Rate in Rs. per meter.	Permitted length in meter.	Paid amount in Rs.	Actually digging length in meter	Amount of digging length in Rs.

While taking trench, we have observed the following lapses :-

No.:	Particular	Penalty in Rs.	Per
1.	Non fixing of barricading at site	1000	Meter
2.	Non displaying information board at site @ 100-meter distance	500	Meter
3.	Not Providing signage boards and reflectory signage's at the site	500	Meter
4.	Non appointing of warden/assistance at crowded traffic road per site	200	Day
5.	After cable laying / service line work complete, non-removal of excavated balance material /soil from site in time per site	1000	Day
6.	Not cleaning storm water /drainage line Chamber if excavated soil fall in chamber	1000	Per chamber
7.	Damage to Municipal utility/Private property/ Other utility	Actual Cost	
8.	Increase in trenching length against sanctioned work order – (More than 5%of approval)	Three times of approved flat road per Meter	

Since you have exceeded the trench length than the permitted therefore you shall make a payment of Rs. _____ within Seven days to the PMC Office.

You are liable to penalties to the amount calculated as above. Hence you have to pay payment of Rs.----- within seven days to the PMC Office.If you fail to make this payment , note that your Bank Guarantee will be forfeited to recover the penalty amount and further permission for trenching activity will not be granted,irrespective of any emergency .

Executive Engineer
Road Department
Pune Municipal Corporation.

Copy to :1.Head Clerk (Road Dept.)
Pune Municipal Corporation.

Annexure -14

PUNE MUNICIPAL CORPORATION
FORM OF REGISTER FOR TRENCHING
(To be Maintained by Head Clerk,Road Department)

Sr. No.	Work Order No.& Date	Name of the Utility	Location	Length on footpath/ Passages	Length on Carriageway	Surface of Carriageway

Amount of Reinstatement Charges	5 % Bank Guarantee received details	Details of Reinstatement Amount (DD No,Bank etc.)	Amount of Recovery/ Refund to Utility	Signature of Clerk	Checked by Accountant

Annexure -15

PUNE MUNICIPAL CORPORATION
LETTER FOR RECTIFICATION TO REINSTATEMENT CONTRACTOR

To:

Sub :- Rectification of settled trench during defect liability period

Ref:- Work Order No. Date.

The above trench was reinstated by you under above referred Work Order onon The inspection was carried out and found that at the reinstated portion is

- 1. Settled -----
- 2. Raised -----
- 3. -----

As per the conditions of contract and this being under defect liability period, you shall re-excavate the complete reinstated portion as above mentioned and reinstate it again within Seven days from the date of receipt of this letter.

It shall be mentioned that if you fail to attend this work within stipulated period you will penalized with Rs.1,000/- per Rmt. and it will be recovered from your security deposit.

Similarly this trench will be reinstated at your risk and cost which you shall note.

Executive Engineer
Road deapartment
Pune Municipal Corporation.

Annexure -16

PUNE MUNICIPAL CORPORATION

Monitoring of Trenching Activity

STATUS OF TRENCHING ACTIVITY

Road Dept.				Report For The Month -			Year-	
Name of the utility agency /dept	Length of the trench permitted during the month (in meters)			Length of the trench excavated in this month (in meters) (out of 4)	Cumulative length of trench excavated since (in meters)	Cumulative length of trench reinstated since (in meters)	Balance length to be reinstated (in meters)	Reasons for delay & no of days of delay if any
	For laying	For repairs	Total (2+3)					

Junior Engineer
Road department
Pune Municipal Corporation.

Deputy Engineer
Road department
Pune Municipal Corporation

Ex.Engineer
Road department
Pune Municipal Corporation

Annexure -17

PUNE MUNICIPAL CORPORATION
RECONCILIATION STATEMENT

Sub. : - Trench on _____

Name of Agency :- -----

Ref. : - 1) Circular No.
2) This office Work Order no._____ dt. _____

The Reinstatement Charges to be recovered are as follows.

No.	Particular	Flat Rate in Rs. per meter.	Permitted length in meter.	Paid amount in Rs.	Actually digging length in meter	Amount of digging length in Rs.

No.:	Particular	Penalty in Rs.	Per
1.	Non fixing of barricading at site	1000	Meter
2.	Non displaying information board at site @ 100-meter distance	500	Meter
3.	Not Providing signage boards and reflectory signage's at the site	500	Meter
4.	Non appointing of warden/assistance at crowded traffic road per site	200	Day
5.	After cable laying / service line work complete, non-removal of excavated balance material /soil from site in time per site	1000	Day
6.	Not cleaning storm water /drainage line Chamber if excavated soil fall in chamber	1000	Per chamber
7.	Damage to Municipal utility/Private property/ Other utility	Actual Cost	
8.	Increase in trenching length against sanctioned work order – (More than 5%of approval)	Three times of approved flat road per Meter	

Amount paid by Utility against reinstatement charges Rs -----

Amount as per actually digging length Rs -----

Penalty Charges as described as above Rs -----

Therefore the amount to be recovered / refundable Rs -----

Junior Engineer
Road department
Pune Municipal Corporation.

Deputy Engineer
Road department
Pune Municipal Corporation

Ex.Engineer
Road department
Pune Municipal Corporation

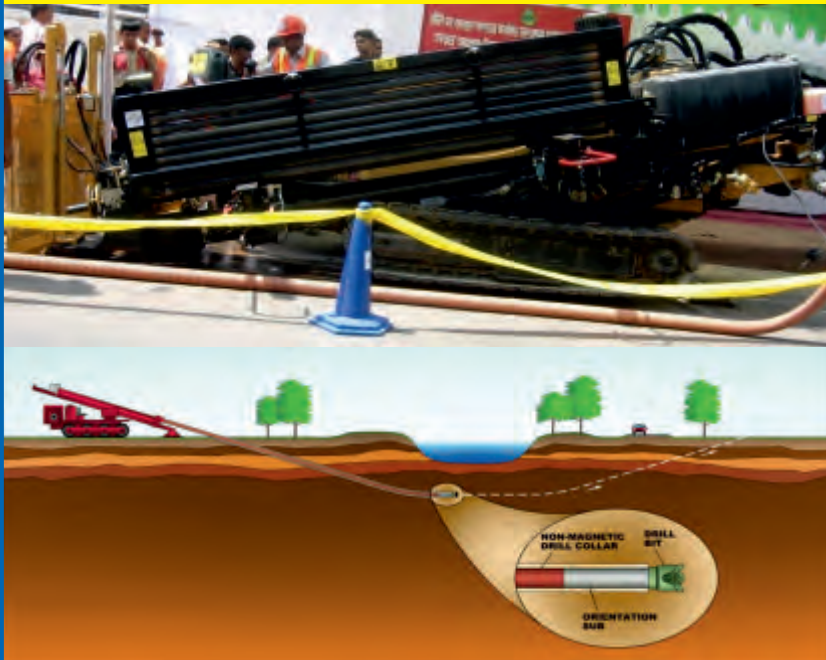
SUB-SURFACE SURVEY BY G.P.R



G.P.R.SURVEY



Horizontal Directional Drilling Machine



Drilling head reached at other end



Drilling head reached at other end



Fitting of reaming tool



Fixing of pulling head



Dragging of pipe



